Learning Standard 1: Ideate

Gr.	Emerging / Emerging+	Developing / Developing+	Proficient / Proficient+	Extending / Extending+
9	Begins to explore research to uncover design opportunities, guided by curiosity and compassion. States design idea, beginning to recognize contextual factors (social, ethical, and/or sustainability).	Explores research to understand some needs and uncover design opportunities. Identifies design ideas, recognizing some contextual factors (social, ethical, and/or sustainability).	Explores research to understand needs and uncover purposeful design opportunities. Outlines design ideas based on intended impact, recognizing several contextual factors.	Explores and describes research to uncover meaningful and purposeful design opportunities. Describes feasible and comprehensive design ideas based on intended impact, recognizing a range of contextual factors.
10	Explores research to uncover design opportunities. Identifies design idea, beginning to recognize contextual factors (social, ethical, and/or sustainability).	Explores and describes research to uncover purposeful design opportunities. Outlines design ideas based on intended impact, recognizing some contextual factors.	Explores and explains research to understand needs and uncover purposeful design opportunities. Describes design ideas based on intended impact, recognizing a range of contextual factors.	Explores and justifies research to uncover exceptionally meaningful design opportunities. Explains feasible and comprehensive design ideas based on intended impact, recognizing a range of contextual factors.
11	Explores and describes research to uncover design opportunities. Outlines design ideas, recognizing some contextual factors (social, ethical, and/or sustainability).	Explores and explains research to uncover purposeful design opportunities. Describes design ideas based on intended impact, recognizing a range of contextual factors.	Explores and justifies research to uncover purposeful design opportunities. Explains design ideas and their limitations, recognizing a range of contextual factors.	Justifies research to uncover exceptionally meaningful design opportunities. Justifies feasible and comprehensive design ideas, explaining limitations and a range of contextual factors.
12	Explores and explains research to uncover purposeful design opportunities. Describes design ideas, describing limitations and some contextual factors.	Explores and justifies research to uncover purposeful design opportunities. Explains design ideas, describing limitations and a range of contextual factors.	Synthesizes and justifies research to uncover purposeful design opportunities. Justifies design ideas, explaining limitations and a range of contextual factors.	Synthesizes and justifies research to uncover exceptionally meaningful design opportunities. Justifies and evaluates comprehensive design ideas, explaining limitations and a wide range of contextual factors.

Learning Standard 2: Prototype

Gr.	Emerging / Emerging+	Developing / Developing+	Proficient / Proficient+	Extending / Extending+
9	Begins to construct a development and production plan, stating a prototyping method, tools, materials, and procedures. Begins to use learning from tests, feedback, and/or data to refine or abandon designs.	Constructs a partial development and production plan, identifying a prototyping method, tools, materials, and procedures. Uses learning from tests, feedback, and/or data to refine or abandon designs.	Constructs a development and production plan, outlining a prototyping method, tools, materials, and procedures. Applies learning from tests, feedback, and/or data to refine or abandon designs.	Thoroughly constructs a development and production plan, describing a prototyping method, tools, materials, and procedures. Analyzes learning from tests, feedback, and/or data to refine or abandon designs.
10	Begins to construct a development and production plan, identifying a prototyping method, tools, materials, and procedures. Begins to apply learning from tests, feedback, and/or data to refine or abandon designs.	Constructs a partial development and production plan, outlining a prototyping method, tools, materials, and procedures. Applies learning from tests, feedback, and/or data to refine or abandon designs.	Constructs a development and production plan, describing a prototyping method, tools, materials, and procedures. Synthesizes learning from tests, feedback, and/or data to refine or abandon designs.	Thoroughly constructs a development and production plan, explaining a prototyping method, tools, materials, and procedures. Thoroughly synthesizes learning from tests, feedback, and/or data to refine or abandon designs.
11	Begins to construct a development and production plan, outlining a prototyping method, tools, materials, and procedures. Begins to synthesize learning from tests, feedback, and/or data to refine or abandon designs.	Constructs a partial development and production plan, describing a prototyping method, tools, materials, and procedures. Synthesizes learning from tests, feedback, and/or data to refine or abandon designs.	Constructs a development and production plan, explaining a prototyping method, tools, materials, and procedures. Evaluates learning from tests, feedback, and/or data to refine or abandon designs.	Thoroughly constructs a development and production plan, justifying a prototyping method, tools, materials, and procedures. Thoroughly evaluates learning from tests, feedback, and/or data to refine or abandon designs.
12	Begins to construct a development and production plan, describing a prototyping method, tools, materials, and procedures. Begins to evaluate learning from tests, feedback, and/or data to refine or abandon designs.	Constructs a partial development and production plan, explaining a prototyping method, tools, materials, and procedures. Evaluates some learning from tests, feedback, and/or data to refine or abandon designs.	Constructs a development and production plan, justifying a prototyping method, tools, materials, and procedures. Evaluates, with explanation, learning from tests, feedback, and/or data to refine or abandon designs.	Constructs a robust development and production plan, justifying a prototyping method, tools, materials, and procedures. Critically evaluates, with justification, learning from tests, feedback, and/or data to refine or abandon designs.

Learning Standard 3: Make

Gr.	Emerging / Emerging+	Developing / Developing+	Proficient / Proficient+	Extending / Extending+
9	Follows a plan with frequent support; may need reminders to describe changes. Selects tools or materials with guidance. Technical execution of the plan is in the early stages.	Follows a plan with some help, sometimes describing simple changes. Selects common tools and materials with some support. Execution of the plan shows growth.	Follows a plan with limited support, describing and making purposeful changes. Selects appropriate tools and materials for tasks and executes the plan competently.	Follows and adapts a plan independently, clearly explaining decisions and changes. Confidently selects and uses tools and materials effectively to execute the plan.
10	Follows a plan with guidance, describing basic changes. Needs confirmation when selecting tools or materials; executes the plan with support.	Follows a plan with some independence, occasionally describing and justifying changes. Begins selecting tools and materials with limited input; execution of the plan is improving.	Follows and adapts a plan independently, clearly describing changes. Selects appropriate tools and materials with growing confidence and executes the plan soundly.	Independently adapts and refines a plan, justifying changes with insight. Selects and applies tools and materials confidently and appropriately; execution is precise and effective.
11	Follows a plan with support, describing changes made. Requires confirmation that the correct tools, technologies, and/or materials are selected for tasks.	Follows a plan with some independence, occasionally describing and justifying changes made. Demonstrates some confidence in selecting appropriate tools, technologies, and/or materials.	Follows and adapts a plan with increasing independence, clearly describing and justifying changes made. Effectively and independently selects suitable tools, technologies, and materials for tasks.	Independently creates, follows, and adapts a detailed plan, thoughtfully explaining and evaluating changes. Leads and/or enhances collaborative efforts, thoughtfully selecting and applying tools, technologies, and materials.
12	Follows a plan with support, explaining changes made. Requires confirmation that the best available tools, technologies, and/or materials are selected for tasks.	Follows a plan with some independence, occasionally describing and justifying changes made. Demonstrates some confidence in selecting appropriate tools, technologies, and/or materials.	Follows and adapts a plan with increasing independence, clearly describing and justifying changes made. Effectively and independently selects suitable tools, technologies, and materials for tasks.	Independently creates, follows, and adapts a detailed plan, thoughtfully explaining, justifying, and evaluating changes. Leads and/or enhances collaborative efforts, expertly selecting and applying tools, technologies, and materials.

Learning Standard 4: Skills

Gr.	Emerging / Emerging+	Developing / Developing+	Proficient / Proficient+	Extending / Extending+
9	Follows safety procedures in physical and digital environments with frequent reminders. Demonstrates early-stage technical skills, requiring step-by-step support.	Follows safety procedures with occasional reminders. Demonstrates developing technical skills on familiar tasks.	Consistently follows safety procedures for self and others. Demonstrates competent technical skills across required tasks.	Consistently applies safety procedures and encourages safe practice in others. Demonstrates strong, careful technical skills. Describes skills needed and sets specific goals to refine them.
10	Follows safety procedures with reminders. Demonstrates emerging technical skills with support. Identifies some skills needed for the task.	Follows safety procedures with growing consistency. Demonstrates developing technical skills with increasing accuracy. Outlines skills needed and practices them.	Consistently follows safety procedures for self and others. Demonstrates sound, consistent technical skills. Describes skills needed for design interests and follows a plan to refine them.	Consistently applies and models safety procedures. Demonstrates precise and effective technical skills. Explains skills needed and develops specific plans to refine them over time.
11	Follows safety procedures with support on increasingly complex tasks. Demonstrates emerging technical skills. Outlines skills needed for design interests.	Follows safety procedures with some independence. Demonstrates developing technical skills on complex tasks. Describes skills needed and begins a plan to refine them.	Independently maintains safe practices for self, co-workers, and users. Demonstrates consistent technical skills on complex tasks. Develops specific plans to learn or refine identified skills.	Models and promotes safe practices in physical and digital environments. Demonstrates refined technical skills involving dexterity and complexity. Evaluates skill growth and adjusts plans to extend skills over time.
12	Follows safety procedures with prompting on complex tasks. Demonstrates emerging technical skills. Describes skills needed for design interests.	Follows safety procedures with some independence. Demonstrates developing technical skills on complex tasks. Explains skills needed and follows a plan to refine them.	Independently maintains and verifies safe practices for self, co-workers, and users. Demonstrates competent, consistent technical skills on complex tasks. Implements and monitors specific plans to refine skills over time.	Leads safe practice in physical and digital environments. Demonstrates refined and advanced technical skills. Critically evaluates skill development and supports the skill growth of others.

Learning Standard 5: Evaluate

Gr.	Emerging / Emerging+	Developing / Developing+	Proficient / Proficient+	Extending / Extending+
9	Presents the product using basic terminology for an intended audience. States how the product served individuals, communities, and/or the environment. With guidance, states how a chosen tool, technique, and/or technology suited the task.	Presents the product with rationale, using some accurate terminology. Identifies how the product served individuals, communities, and/or the environment. Identifies how chosen tools, techniques and/or technologies suited the task.	Presents the product with rationale, using accurate terminology for an intended audience. Outlines how the product served individuals, communities, and/or the environment. Outlines how well chosen tools, techniques and/or technologies suited the task, and their impacts.	Presents the product with rationale, using a range of precise terminology. Describes how the product served individuals, communities, and/or the environment. Describes impacts of tools, techniques, and technology choices, including unintended consequences.
10	Presents the product using basic terminology for an intended audience. Identifies how the product served individuals, communities, and/or the environment. Identifies how chosen tools, techniques and/or technologies suited the task.	Presents the product with rationale, using some accurate terminology. Outlines how the product served individuals, communities, and/or the environment. Outlines how well chosen tools, techniques and/or technologies suited the task, and their impacts.	Presents the product with rationale, using accurate terminology for an intended audience. Describes how the product served individuals, communities, and/or the environment. Describes impacts of tools, techniques, and technology choices, including unintended consequences.	Presents the product with rationale, using a range of precise terminology. Explains how the product served individuals, communities, and/or the environment. Explains impacts of technology choices, including unintended consequences.
11	Presents the product using basic terminology for an intended audience. Outlines how the product served individuals, communities, and/or the environment. Outlines how chosen tools and technologies suited the task.	Presents the product with rationale, using accurate terminology. Describes how the product served individuals, communities, and/or the environment. Describes impacts of tools, techniques, and technology choices, including unintended consequences.	Presents the product with rationale, using accurate terminology for an intended audience. Explains and evaluates how the product served individuals, communities, and/or the environment. Explains impacts of tools, techniques and/or technology choices, including unintended consequences.	Presents the product with rationale, using a range of precise terminology. Justifies and evaluates how the product served individuals, communities, and/or the environment. Evaluates the suitability and impacts of tools, techniques, and/or technology choices and how others might build on the design.
12	Presents the product using basic terminology for an intended audience. Describes how the product served individuals, communities, and/or the environment. Describes impacts of tools, techniques, and technology choices.	Presents the product with rationale, using accurate terminology. Explains and evaluates how the product served individuals, communities, and/or the environment. Explains impacts of tools, techniques and/or technology choices, including unintended consequences.	Presents the product with rationale, using accurate terminology for an intended audience. Justifies and evaluates how the product served individuals, communities, and/or the environment. Evaluates the suitability and impacts of tools, techniques, and/or technology choices.	Presents the product with rationale, using a range of precise terminology. Critically evaluates and justifies how the product served individuals, communities, and/or the environment. Evaluates the role of changing technologies in the discipline and the impacts of technology choices.